# Answers on
# Booklet - 11

# Taken from OCR Physics A past papers

| | | | |
|---|---|---|---|
| Abbreviations, annotations and conventions used in the Mark Scheme | / | = | alternative and acceptable answers for the same marking point |
| | ; | = | separates marking points |
| | NOT | = | answers which are not worthy of credit |
| | ( ) | = | words which are not essential to gain credit |
| | ____ | = | (underlining) key words which **must** be used to gain credit |
| | ecf | = | error carried forward |

1.

(a) (i) amplitude correctly labelled (by **A** or in words ) ---------- B1 [1]
(reject "A" as a point i.e. with no arrows)

(ii) wavelength correctly labelled (by λ or in words) ---------- B1 [1]

(b) (i) same shape ---------- B1
moved slightly to the right consistently drawn for both waves ---------- B1 [2]
(do not allow shift of more than ¼ wavelength)

(ii) movement is VERTICAL ---------- M1
Q moves UP ↑ <u>AND</u> S moves DOWN ↓ shown ---------- A1 [2]

(c) phase difference = 180° (degrees) OR π ---------- B1 [1]
{allow "in antiphase" do not allow "out of phase"}

(d) (i) recall of T = 1/f ---------- C1
T = 1/25 = 0.04 s ---------- A1 [2]

(ii) recall of v= fλ ---------- C1
valid substitution: e.g. v = 25 x .036 ---------- C1
v = 0.90 $ms^{-1}$ ---------- A1 [3]
(there are 2 possible errors – incorrect wavelength and wrong units, so
v = 90 m/s scores 2 marks
v = 0.45 m/s scores 2 marks but allow 3 marks for ecf from cand's λ in (a) (ii)
v = 45 m/s scores 1 mark but allow 2 marks for ecf from cand's λ in (a) (ii)

2.

| | | | | |
|---|---|---|---|---|
| **(a)** | **(i)**<br>**(ii)** | diffraction or refraction or superposition or interference<br>only transverse waves can be polarised | B2<br>B1 | **accept** any two from the four listed<br>**accept** sound is a longitudinal wave or e-m waves are transverse |
| | **(iii)** | place transmitter and receiver facing each other<br><br>rotate either transmitter or receiver through 90° about axis joining aerials<br>**or** use two polarising filters and rotate from parallel to crossed<br><br>observe signal fall to zero/minimum from initial high value on meter<br>monitoring output of receiver<br>explanation of observations/link between observations and polarisation | B1<br><br>B1<br><br><br><br>B1<br>B1 | **accept** from diagram<br>**allow** (metal) grille/polarising filter to polarise microwaves<br>**accept** place (metal) grille/polarising filter [**not** Polaroid] between transmitter and receiver and rotate through 90°<br><br><br>QWC mark |

3.

| | | |
|---|---|---|
| **(a)(i)** | Electron | B1 |
| **(ii)** | Ion | B1 |
| **(b)** | Voltage / p.d. | B1 |
| | Current | B1 |
| | (R = V/I scores 1/2 if symbols are not defined) | |
| | *(Symbols 'V' and 'I' alone score 0/2)* | |
| **(c)(i)** | $I = \Delta Q/\Delta t$ (Allow other variants - Δ not essential ) | C1 |
| | $I = 650 / 1.6 \times 10^{7}$ | C1 |
| | $I = 4.0(6) \times 10^{-5}$ (A) | A1 |
| **(ii)** | number $= 650/1.6 \times 10^{-19}$ | C1 |
| | number $= 4.0(6) \times 10^{21}$ | A1 |

4.

| | | |
|---|---|---|
| (a) | The energy transformed by a 1 kW device in a time of 1 hour | B1 |
| (b)(i)1. | time = 4.0 × 7 = 28 (hours) / power = 0.11 (kW) | C1 |
| | number of kW h = 0.110 × 28 | |
| | number of kW h = 3.08 ≈ 3.1 (If 4 hours used, them 0.44 scores 1/2) | A1 |
| (b)(i)2. | cost = 3.08 × 7.5 (Possible ecf) | |
| | cost = 23 (p) | B1 |

5.

| | |
|---|---|
| $E = Pt$ / 5.8 = 0.12 × time / (time =) 48.3 (hr) | C1 |
| (time =) $1.74 \times 10^{5} \approx 1.7 \times 10^{5}$ (s) | A1 |

6.

| | | | | |
|---|---|---|---|---|
| **a** | **i** | Correct symbol for thermistor | B1 | To score the symbol mark, must have an oblong with a 'line + bend at one end' through the oblong in **any** orientation (do not allow arrow) |
| | | The resistance decreases as the temperature (of thermistor) increases (ora) | B1 | **Allow** It increases as the temperature decreases (1)<br>**Not** any reference to 'heat' instead of temperature |
| | **ii** | Correct symbol for LDR | B1 | To score the symbol mark, must have an oblong plus a minimum of one arrow towards the oblong – the circle round the oblong is not necessary. |
| | | The resistance decreases as intensity / light increases (ora) | B1 | **Allow** In light the resistance is low / small (1)<br>**Allow** In dark(ness) the resistance is high / large / big (1) |
| **b** | **i** | A line / curve of positive gradient | B1 | To score this mark, the line or curve must have a finite resistance value at 0 °C - allow resistance intercept at or above the 'top of zero' on the l.h.s.<br>**Not** A line/curve that becomes horizontal |
| **ii** | | | | **Must show ticks on the script to indicate where marks are being awarded** |
| | | Place the wire in a (water) bath / use a hot plate / oven | B1 | **Not** 'heat the wire' |
| | | Any remaining four from:<br>1. Connect conductor / wire in series with a battery / cell / power supply<br>2. Ammeter placed in series (with wire) / voltmeter placed in parallel (with wire)<br>3. Record / read / measure current / voltage (across the wire) (AW)<br>4. (Calculate the resistance using the equation:) $R = \frac{V}{I}$<br>5. Thermometer / 'temperature probe' mentioned<br>6. Measurements (for *V* and *I* ) repeated for different temperatures | B1 × 4 | Numbered marking points 1 and 2 can be scored on diagram or in the text but **all** others marking points must be written<br>Not '*V* = *IR*' for numbered marking point 4<br>**Not** '*Plot a graph of R against temperature*' for numbered marking point 6 |

7.

| | | |
|---|---|---|
| (a) | $R = V / I$ | C1 |
| | symbols defined: $R$ = resistance, V = p.d. / voltage and $I$ = current | A1 |
| | ( $V = IR$ with all symbols defined scores 1/ 2) | |
| | ( $R$ = p.d. / voltage per unit current scores 2/2) | |
| | ( $R$ = p.d. / voltage per unit amp / A scores 1/ 2) | |
| (b) | Resistance decreases as temperature increases. | B1 |
| | Correct curve with $R$ decreasing as temperature increases. | B1 |
| (c)(i) | Resistance increases (as $V$ increases) | B1 |
| | Temperature increases / atoms vibrate more / more electron collisions (with atoms) | B1 |
| (ii) | **1.** $P = 24 \times 2$ | C1 |
| | $P = 48$ (W) | A1 |
| | **2.** $V = 12$ (V) when current is 2.0 (A) (Allow $V = 11.0$ V from graph) | C1 |
| | $R = 12 / 2.0 = 6.0$ (Ω) | A1 |

8.

Any **four** marks from:

| | | |
|---|---|---|
| Resistance of thermistor decreases as temperature increases | (ora) | **B2** |
| (Resistance of circuit changes with temperature scores 1 mark) | | |
| The voltmeter reading / voltage stays constant / at 4.5 (V) | (AW) | **B1** |
| The ammeter reading / current increases | | **B1** |
| The current is inversely proportional to the resistance of the thermistor | | **B1** |

9.

| | | |
|---|---|---|
| (a) | Electron flow is in opposite direction (to conventional current) | B1 |
| (b) | Correct symbol for the LDR | B1 |
| | (Resistance of LDR) decreases with increased intensity / brightness / light (AW) | B1 |
| (c) | current ∝ p.d. (Allow 'voltage' instead of p.d.) | B1 |
| | (provided the) temperature (of metallic conductor) remains constant | B1 |
| | (voltage = current × resistance scores 0/2) | |
| | ($V = IR$ and $R$ = constant scores 0/2) | |
| (d)(i)1. | $R$-$V$ graph for metallic conductor: | |
| | shows $R$ = constant / 'horizontal line' | B1 |
| (d)(i)2. | $R$-$V$ graph for thermistor: | |
| | shows $R$ has a <u>finite</u> value at $V = 0$ | B1 |
| | shows $R$ decreases as $V$ increases (Allow a 'curve' or 'straight line') | B1 |
| (d)(ii)1. | Any <u>two</u> from: | |
| | The resistances larger / line (graph) higher (and horizontal) (Can score on Fig.1.2 a) | |
| | The electrons collide more often / frequently (with vibrating atoms) | |
| | The atoms / ions vibrate 'more' (Do not allow '*particles*' vibrate) | B1 × 2 |
| (d)(ii)2. | The resistance increases / doubles (Can be scored on Fig.1.2a) | M1 |
| | Mention of: $R \propto L$ or $R = \frac{\rho l}{A}$. | A1 |

10.

| | | |
|---|---|---|
| (a)(i) | 1. *constructive:* waves meet in phase (WTTE) | B1 |
| | producing a resultant of maximum amplitude (WTTE) | B1 |
| | *{allow producing "a bright spot" or "a loud sound"}* | |
| | 2. *destructive:* waves meet in antiphase OR out of phase | B1 |
| | *{allow "waves cancel each other out"}* | |
| | producing a resultant of minimum amplitude (WTTE) | B1 |
| | *{allow producing "a black spot" OR "no light" OR "no sound"}* | |
| | | [4] |
| | *{Full marks may be scored for correct and fully labelled diagrams}* | |
| (ii) | *coherence*: constant phase difference | B2 |
| | *{allow a B1 mark for "same frequency" OR "identical waves"}* | [2] |
| (b)(i) | *maxima:* where path difference = whole number of wavelengths | B1 |
| | *minima:* where path difference = odd number of half-wavelengths | B1 [2] |
| (ii) | required path difference for maxima/minima now shorter | B1 |
| | hence separation would DECREASE | B1 [2] |

11.

***diagram***:

laser OR light source <u>and single-slit</u> in front of double slit ---------- B1

screen (WTTE) (or travelling microscope) behind double-slit ---------- B1

(if 'screen' is not labelled mark can be obtained by reference to 'screen' in text)

***measurements:***

<u>measure</u> distance between double-slit and screen ---------- B1

<u>measure</u> distance between neighbouring dark/bright images ---------- B1

(allow 'fringe spacing' or measure distance for n fringes)

***formula:***

recall of $\lambda = ax/D$ ---------- B1

***ALL symbols correctly defined***

a = distance between slits ----------}

x = fringe separation (WTTE) ----------} B1

D = distance between slits and screen ----------} [6]

(If candidate uses their own symbols they must be used correctly to score the formula recall mark)

(do not penalise careless use of d and D: i.e. being interposed)

12.

| | | | |
|---|---|---|---|
| (a) | when two waves **meet** (OR interfere) (at a point) | B1 | |
| | the resultant displacement is **sum** of individual displacements (WTTE) | B1 | [2] |
| | ***{allow 'amplitude'}*** | | |
| (b)(i) | same frequency OR wavelength | B1 | |
| | "in phase" (WTTE) | B1 | [2] |
| | ***{n.b. "constant phase difference scores 2 marks}*** | | |
| (ii) | (path difference =) $S_2P - S_1P = (½)\lambda$ | B1 | [1] |
| (iii) | evidence of realisation that "fringe separation" $x$ = 8mm | B1 | |
| | recall of $\lambda = ax/D$ (allow candidate's choice of symbol for x) | C1 | |
| | $a = \lambda D/x = (6.4\times10^{-7} \times 1.5)/8 \times 10^{-3} =$ **$1.2 \times 10^{-4}$ m** | A1 | [3] |
| | ***{Allow ecf for cand's x: eg if x = 4mm, a = 2.4 x $10^{-4}$ m - this scores 2 marks}*** | | |
| (iv) | maximum intensity when y=0 AND minima at +4 and –4 | B1 | |
| | **correct 'period' i.e 8 mm** with at least 2 full cycles drawn | B1 | [2] |